US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12397471
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Wang; Zhigang et al.

Ultrasonic machining an aperture in a workpiece

Abstract

A method is provided for machining a workpiece. During this machining method, an aperture is formed in the workpiece using a machining system. The machining system includes an ultrasonic machining device, a slurry delivery device and a controller. The forming of the aperture includes delivering a slurry to an interface between the ultrasonic machining device and the workpiece using the slurry delivery device, and transmitting ultrasonic vibrations into the slurry using the ultrasonic machining device. A feedback parameter is monitored during the forming of the aperture using the controller. A slurry delivery parameter for the slurry delivery device is adjusted during the forming of the aperture based on the feedback parameter using the controller.

Inventors: **Wang; Zhigang (South Windsor, CT), Riehl; John D. (Hebron, CT), Fernandez; Robin H. (East Haddam, FL), Kuczek; Andrzej E. (Bristol, CT), Srinivasan; Gajawalli V. (South Windsor, CT)**

Applicant: **Raytheon Technologies Corporation** (Farmington, CT)

Family ID: **1000008777367**

Assignee: **RTX Corporation (Farmington, CT)**

Appl. No.: **17/554748**

Filed: **December 17, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20230191658 A1	Jun. 22, 2023

Publication Classification

Int. Cl.: **B28B11/00** (20060101); **B28B11/08** (20060101); **B28B13/02** (20060101); **B28B17/00** (20060101)

U.S. Cl.:

CPC **B28B11/0881** (20130101); **B28B13/0275** (20130101); **B28B17/0081** (20130101); B28B2013/0265 (20130101)

Field of Classification Search

CPC: B28B (11/0881); B28B (13/0275)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4754115	12/1987	Rhoades	219/69.15	B24B 47/20
4828052	12/1988	Duran	N/A	N/A
4926309	12/1989	Wu	706/904	G05B 19/4163
6458225	12/2001	Statnikov	188/73.1	C21D 10/00
2001/0027354	12/2000	Kakino	700/173	G05B 19/4163
2019/0143431	12/2018	Luo	N/A	N/A
2020/0324346	12/2019	Whittle	N/A	B23B 37/00
2021/0260713	12/2020	Song	N/A	N/A
2022/0161387	12/2021	Yeo	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
S63103941	12/1987	JP	N/A
2021100705	12/2020	WO	N/A

OTHER PUBLICATIONS

Dhinakaran, V., Katiyar, J.K. and Jagadeesha, T., 2021. Mechanics and Material Removal Modeling and Design of Velocity Transformers in Ultrasonic Machining. In Mechatronic Systems Design and Solid Materials (pp. 99-146). Apple Academic Press. (Year: 2021). cited by examiner
Goswami, D. and Chakraborty, S., 2015. Parametric optimization of ultrasonic machining process using gravitational search and fireworks algorithms. Ain Shams Engineering Journal, 6(1), pp. 315-331. (Year: 2015). cited by examiner
Kumar, J., 2013. Ultrasonic machining—a comprehensive review. Machining Science and Technology, 17(3), pp. 325-379. (Year: 13). cited by examiner
EP search report for EP22213177.3 dated Nov. 24, 2023. cited by applicant

---

Primary Examiner: Krasnow; Nicholas R

Attorney, Agent or Firm: Getz Balich LLC

Background/Summary

BACKGROUND OF THE DISCLOSURE
1. Technical Field
(1) This disclosure relates generally to machining and, more particularly, to ultrasonic machining.

2. Background Information

(2) Ultrasonic machining may be used to form an aperture in a workpiece. Various systems and method for ultrasonic machining are known in the art. While these known ultrasonic machining systems and methods have various benefits, there is still room in the art for improvement. For example, during known methods, material removal rate may slow and a tool tip may wear down quickly from constant impact of abrasive particles due to micro erosion mechanisms during ultrasonic machining of deep apertures. There is a need in the art therefore for improved system and method for ultrasonic machining deep apertures in a workpiece.

SUMMARY OF THE DISCLOSURE

(3) According to an aspect of the present disclosure, a method is provided for machining a workpiece. During this machining method, an aperture is formed in the workpiece using a machining system. The machining system includes an ultrasonic machining device, a slurry delivery device and a controller. The forming of the aperture includes delivering a slurry to an interface between the ultrasonic machining device and the workpiece using the slurry delivery device, and transmitting ultrasonic vibrations into the slurry using the ultrasonic machining device. A feedback parameter is monitored during the forming of the aperture using the controller. A slurry delivery parameter for the slurry delivery device is adjusted during the forming of the aperture based on the feedback parameter using the controller.

(4) According to another aspect of the present disclosure, another method is provided for machining a workpiece. During this machining method, a slurry is delivered to an interface between an ultrasonic machining device and the workpiece. Ultrasonic vibrations are transmitted into the slurry at the interface using the ultrasonic machining device to form an aperture in the workpiece. The slurry and debris from the forming of the aperture are extracted through a passage that extends within the ultrasonic machining device to a tip of the ultrasonic machining device.

(5) According to still another aspect of the present disclosure, a machining system is provided for forming an aperture in a workpiece. The machining system includes a slurry delivery device, an ultrasonic machining device and a controller. The slurry delivery device is configured to deliver a slurry to an interface between the ultrasonic machining device and the workpiece. The ultrasonic machining device is configured to transmit ultrasonic vibrations into the slurry at the interface to form the aperture in the workpiece. The controller configured to: monitor a feedback parameter during the forming of the aperture; provide a control signal based on the feedback parameter; and communicate the control signal to the slurry delivery device to adjust a parameter of the delivery of the slurry to the interface.

(6) The slurry and the debris may be drawn from the interface into the passage using a vacuum.

(7) The method may also include: monitoring a feedback parameter during the forming of the aperture; and adjusting a slurry delivery parameter for the delivery of the slurry to the interface during the forming of the aperture based on the feedback parameter.

(8) The workpiece may be configured from or otherwise include a ceramic matrix composite material.

(9) The slurry may include a plurality of abrasive particles within a carrier liquid.

(10) The plurality of abrasive particles may be configured from or otherwise include a carbide and/or diamond.

(11) The slurry delivery parameter may be a pressure of the slurry.

(12) The slurry delivery parameter may be a flowrate of the slurry.

(13) The adjusting of the slurry delivery parameter may initiate flushing out of the slurry at the interface by directing the slurry through the ultrasonic machining device.

(14) The slurry may be pumped through the ultrasonic machining device to the interface.

(15) The slurry may be drawn out from the interface into the ultrasonic machining device.

(16) The feedback parameter may be a load on the ultrasonic machining device.

(17) The feedback parameter may be a forming rate of the aperture.

(18) The feedback parameter may be a size of a tool of the ultrasonic machining device.
(19) The slurry delivery parameter may be adjusted based on a physics-based model.
(20) The slurry delivery device may include a passage that extends within the ultrasonic machining device to a tip of the ultrasonic machining device. The slurry may be delivered to the interface through the passage during the forming of the aperture.
(21) The slurry delivery device may include a passage that extends within the ultrasonic machining device to a tip of the ultrasonic machining device. The slurry may be removed from the interface through the passage during the forming of the aperture.
(22) The workpiece may be configured as or otherwise include a component of a gas turbine engine.
(23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
(24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic illustration of a machining system.
(2) FIG. **2** is a schematic illustration of an interface between an ultrasonic machining tool and a workpiece during transmission of ultrasonic vibrations.
(3) FIG. **3** is an illustration of the ultrasonic machining tool.
(4) FIG. **4** is a flow diagram of a method for forming an aperture in the workpiece.
(5) FIG. **5** is a flow diagram of a method for controlling ultrasonic machining.
(6) FIGS. **6**A-B are schematic illustrations depicting a sequence for flushing a partially formed aperture.
(7) FIG. **7** is a sectional illustration of the ultrasonic machining tool configured with an internal passage.
(8) FIG. **8** is an enlarged partial sectional illustration of the ultrasonic machining tool with the internal passage fluidly coupled with another component of the machining system.
(9) FIG. **9**A is a partial sectional illustration depicting the internal passage of the ultrasonic machining tool directing slurry to a tool-workpiece interface.
(10) FIG. **9**B is a partial sectional illustration depicting the internal passage of the ultrasonic machining tool extracting slurry from the tool-workpiece interface.
DETAILED DESCRIPTION
(11) FIG. **1** illustrates a machining system **20** for forming and, more particularly, ultrasonic machining of an aperture **22** in a workpiece **24**. This machining system **20** includes a workpiece support **26**, a slurry delivery device **27** and an ultrasonic machining device **28**.
(12) The workpiece support **26** is configured to support the workpiece **24** during the forming of the aperture **22**. The workpiece support **26** of FIG. **1**, for example, includes a support surface **30** on which the workpiece **24** may be placed. This workpiece support **26** also includes a support fixture **32** configured to hold (e.g., temporally fix) a position and orientation of the workpiece **24** during the forming of the aperture **22**.
(13) The slurry delivery device **27** is configured to deliver a liquid slurry to an interface **34** at a gap **35** between an ultrasonic machining tool **36** (e.g., a bit) of the ultrasonic machining device **28** and a location on the workpiece **24** where the aperture **22** is to be formed. The slurry delivery device **27** of FIG. **1**, for example, includes a slurry source **38** and at least one slurry nozzle **40**. The source **38** may include a slurry reservoir **42** and a slurry flow regulator **44**. The reservoir **42** is configured to contain a quantity of the slurry before, during and/or after the forming of the aperture **22**. The

reservoir **42**, for example, may be configured as a tank, a cylinder, a pressure vessel or any other container. The flow regulator **44** is configured to direct a regulated flow of the slurry from the reservoir **42** to the nozzle **40**. The flow regulator **44**, for example, may be configured as, or may otherwise include, a pump and/or a valve assembly. The nozzle **40** is configured to direct the slurry received from the source **38** (e.g., the flow regulator **44**) as a flow (e.g., a stream, a jet, etc.) towards/to the tool-workpiece interface **34**; e.g., into the gap **35**.

(14) The slurry delivery device **27** may continuously (or intermittently) provide the slurry to the tool-workpiece interface **34** during the forming of the aperture **22**. By providing the slurry to the tool-workpiece interface **34** throughout the forming of the aperture **22**, the slurry delivery device **27** may displace previously used slurry at the tool-workpiece interface **34** with fresh slurry from the source **38**. This at least partial (or complete) replacement of the slurry at the tool-workpiece interface **34** may remove debris generated as a byproduct from the forming of the aperture **22**, where the debris may be carried away with the displaced used slurry. The slurry delivery device **27** is therefore also configured to remove the debris from the tool-workpiece interface **34**.

(15) The slurry includes a plurality of abrasive particles suspended within and/or otherwise carried by a carrier liquid. The abrasive particles may include carbide particles such as silicon carbide particles and/or boron carbide particles or diamond particles. Examples of the carrier liquid may include water and/or oil.

(16) The ultrasonic machining device **28** is configured to generate ultrasonic vibrations (e.g., vibrations with a frequency equal to or greater than 20 kHz) and transmit those ultrasonic vibrations via sound waves into the slurry at the tool-workpiece interface **34**. Referring to FIG. **2**, the ultrasonic vibrations **46** excite movement of the abrasive particles **48** within the slurry **50** at the tool-workpiece interface **34**, which may cause at least some of the abrasive particles **48** to repetitively contact (e.g., impinge against, strike, etc.) the workpiece **24**. The repetitive contact between the abrasive particles **48** and the workpiece **24** may form microfractures in the workpiece material at the tool-workpiece interface **34** and thereby erode (e.g., machine away) the workpiece material. The ultrasonic machining device **28** is therefore configured to form (e.g., machine) the aperture **22** in the workpiece **24** at the tool-workpiece interface **34**.

(17) The ultrasonic machining device **28** of FIG. **1** includes a tool holder **52** (e.g., a spindle, a chuck, etc.) and the machining tool **36**. The tool holder **52** is configured to support and hold the machining tool **36**. The tool holder **52** may also be configured to position the machining tool **36** relative to the workpiece **24**. The tool holder **52**, for example, may be configured as or otherwise included as part of a robot manipulator or a support fixture.

(18) Referring to FIG. **3**, the machining tool **36** extends along a longitudinal centerline **54** between a back end **56** of the machining tool **36** and a tip **58** at a front end **60** of the machining tool **36**. This machining tool **36** of FIG. **3** includes a tool mount **62**, a tool back mass **64**, a tool transducer **66**, a tool front mass **68**, a tool horn **70** and a tool head **72**. The tool mount **62** is arranged at the tool back end **56** and is configured to mate with and attach to the tool holder **52** of FIG. **1**. The tool back mass **64** is arranged longitudinally between and is connected to the tool mount **62** and the tool transducer **66**. The tool transducer **66** is arranged longitudinally between and is connected to the tool back mass **64** and the tool front mass **68**. This tool transducer **66** is configured to generate the ultrasonic vibrations within the machining tool **36**. The tool front mass **68** is arranged longitudinally between and is connected to the tool transducer **66** and the tool horn **70**. The tool horn **70** is arranged longitudinally between and is connected to the tool front mass **68** and the tool head **72**. This tool horn **70** is configured with a tapered geometry to amplify a vibrational amplitude of the ultrasonic vibrations communicated through the machining tool **36** from the tool transducer **66** to the tool head **72**. The tool head **72** is arranged at the tool front end **60** and projects longitudinally to the tool tip **58**. This tool head **72** of FIG. **2** is configured as a transmitter for transmitting the amplified ultrasonic vibrations **46** into the slurry **50** at the tool-workpiece interface **34**.

(19) FIG. **4** is a flow diagram of a method **400** for forming (e.g., ultrasonic machining) the aperture **22** in the workpiece **24**. The aperture **22** may be a perforation, a through-hole, a recess, a channel, a notch, an indentation or any other type of volume extending partially into or through the workpiece **24**. The workpiece **24** may be constructed from a hard and/or brittle material such as a ceramic; e.g., a pure ceramic material, a ceramic matric composite material, etc. The workpiece **24** may be configured as or included as part of a component for a gas turbine engine, examples of which may include an airfoil, a platform, a shroud, a blade outer air seal (BOAS), a liner and a flowpath wall. The method **400** of the present disclosure, however, is not limited to gas turbine engine workpiece applications. Furthermore, while the method **400** is described below with reference to the machining system **20** described above, the method **400** may alternatively be performed with other machining system arrangements.
(20) In step **402**, the workpiece **24** is positioned with the workpiece support **26**.
(21) In step **404**, the aperture **22** is formed in the workpiece **24**. The slurry delivery device **27**, for example, directs a flow of the slurry to the tool-workpiece interface **34** through, for example, the nozzle **40**. This flow of the slurry may maintain a minimum quantity of the slurry at the tool-workpiece interface **34** such that the gap **35** between the tool tip **58** and the workpiece **24** remains full of the slurry. The flow of the slurry may also maintain a flow (e.g., a current) of the slurry into, through and out of the gap **35** between the tool tip **58** and the workpiece **24**. While this slurry is present at, and/or flowing through, the tool-workpiece interface **34**, the machining tool **36** generates the ultrasonic vibrations and transmits those ultrasonic vibrations into the slurry at the tool-workpiece interface **34** towards the workpiece **24**. These ultrasonic vibrations excite movement of the abrasive particles within the slurry such that at least some of the abrasive particles repetitively contact and vibrate against the workpiece **24** at the tool-workpiece interface **34**. This vibratory contact between the abrasive particles and the workpiece **24** may form microfractures in the workpiece material and erode away the workpiece material at the tool-workpiece interface **34**. The aperture **22** may thereby be formed (e.g., machined) at the tool-workpiece interface **34** in the workpiece **24**.
(22) A formation rate (e.g., machining speed) of the aperture **22** into the workpiece **24** may depend on various parameters. These parameters may include, but not limited to: Amplitude of the ultrasonic vibrations at the tool-workpiece interface **34**; Static pressure of the slurry at the tool-workpiece interface **34**; Concentration of the abrasive particles within the slurry at the tool-workpiece interface **34**; and Size and distribution of the abrasive particles within the slurry at the tool-workpiece interface **34**.
Ideally, where these parameters are maintained substantially constant, the aperture formation rate (e.g., machining speed) should remain substantially constant independent of penetration depth of the tool head **72** into the workpiece **24**; e.g., a measure of how far the tool head **72** projects into the aperture being formed, which may correspond to aperture depth. However, the aperture formation rate in practice may decrease as the tool penetration depth (e.g., the aperture depth) increases. The aperture formation rate may even approach a zero value (e.g., zero speed) as the tool penetration depth approaches a critical value. This critical value may be about ten millimeters (10 mm); however, the specific value may vary based on other aperture characteristics (e.g., diameter, geometry, etc.) and/or material characteristics (e.g., workpiece hardness, etc.).
(23) A decrease in the formation rate may be caused at least in part to a decrease in a concentration of the abrasive particles in the gap **35** between the tool tip **58** and the workpiece **24** at the tool-workpiece interface **34**. For example, as the tool penetration depth (e.g., the aperture depth) increases, it may be more difficult for the fresh slurry to flow into the partially formed aperture as well as more difficult for the used slurry with the debris to flow out of the partially formed aperture. In addition, as the same abrasive particles remain in the gap **35** between the tool tip **58** and the workpiece **24** at the tool-workpiece interface **34**, those abrasive particles may decrease in size, become dull and/or otherwise wear. The worn abrasive particles may thereby become less

efficient at machining away the workpiece material.
(24) To mitigate or prevent the reduction of the aperture formation rate as the tool penetration depth (e.g., the aperture depth) increases, the machining system **20** of FIG. **1** includes a control system **74** (e.g., an operating system) which may implement (e.g., closed-loop) feedback control during the aperture formation method **400**.
(25) The control system **74** is configured to monitor one or more feedback parameters for the machining system **20** during machining system operation and, in particular, during the forming of the aperture **22** in the workpiece **24**. The control system **74** is also configured to provide control signals to one or more components **27** and **28** of the machining system **20** in order to control operation of one or more of those machining system components **27** and **28**. At least some of these control signals may be generated based on the monitored feedback parameters. The control system **74** may thereby implement feedback control of the machining system **20** and its components **27** and **28**. The control system **74** of FIG. **1**, for example, includes a sensor system **76** and a controller **78**.
(26) The sensor system **76** is configured to sense one or more operational characteristics; e.g., variables, values, etc. These operational characteristics may include or may be indicative of the feedback parameters. Examples of the feedback parameters may include: Load (e.g., pressure) applied between the machining tool **36** and the tool holder **52**; Amplitude of the ultrasonic vibrations generated by the tool transducer **66** and/or transmitted by the tool head **72**; Frequency of the ultrasonic vibrations generated by the tool transducer **66** and/or transmitted by the tool head **72**; Spatial position (e.g., vertical position, alignment, etc.) of the machining tool **36** (e.g., the tool head **72**, the tool tip **58**, etc.) relative to a reference (e.g., the workpiece **24**, the workpiece support **26**, etc.); Rate (e.g., speed) of machining tool longitudinal movement (e.g., penetration into the workpiece **24**); Fluid pressure of the slurry at the tool-workpiece interface **34**; Fluid flowrate of the slurry through the tool-workpiece interface **34**; Fluid pressure of the slurry provided to, flowing through, and/or directed out of the nozzle **40**; Fluid flowrate of the slurry provided to, flowing through, and/or directed out of the nozzle **40**; and/or Size of the tool head **72** (e.g., longitudinal length **80** of the tool head **72** of FIG. **3**, lateral width **82** (e.g., diameter) of the tool head **72**, etc.). The sensor system **76** is further configured to communicate sensor data indicative of the operational characteristics and/or the feedback parameters to the controller **78**.
(27) The sensor system **76** may include one or more sensors **84**. Examples of these sensors **84** include, but are not limited to, a pressure sensor, a force sensor, a flow meter, a position sensor and a dimension measurement device.
(28) The controller **78** is configured to generate and provide the control signals to the machining system components **27**, **28** and **76**. Some of these control signals may be generated using (e.g., closed-loop) feedback control logic. For example, controller **78** may monitor one or more of the feedback parameters to determine the (e.g., real time) formation rate of the aperture **22**. Where the aperture formation rate is equal to or less then a threshold, the controller **78** may signal one or more of the machining system components **27** and **28** to adjust an operational parameter. This process may be repeated until the aperture formation rate rises above the threshold and/or another one or more thresholds are met.
(29) The controller **78** may be implemented with a combination of hardware and software. The hardware may include memory **86** and at least one processing device **88**, which processing device **88** may include one or more single-core and/or multi-core processors. The hardware may also or alternatively include analog and/or digital circuitry other than that described above.
(30) The memory **86** is configured to store software (e.g., program instructions) for execution by the processing device **88**, which software execution may control and/or facilitate performance of one or more operations such as those described in the methods below. The memory **86** may be a non-transitory computer readable medium. For example, the memory **86** may be configured as or include a volatile memory and/or a nonvolatile memory. Examples of a volatile memory may include a random access memory (RAM) such as a dynamic random access memory (DRAM), a

static random access memory (SRAM), a synchronous dynamic random access memory (SDRAM), a video random access memory (VRAM), etc. Examples of a nonvolatile memory may include a read only memory (ROM), an electrically erasable programmable read-only memory (EEPROM), a computer hard drive, etc.

(31) FIG. **5** is a flow diagram of a method **500** for controlling ultrasonic machining of the aperture **22**. For ease of description, this control method **500** is described below with reference to the machining system **20**. The method **500**, however, may also be used for various other machining system configurations.

(32) In step **502**, one or more of the feedback parameters are determined. The sensor system **76**, for example, may sense one or more of the operational characteristics and generate sensor data indicative of/based on the sensed operational characteristics. This sensor data is then communicated to the controller **78**. This sensor data may include or be indicative of the feedback parameters. Where the sensor data is indicative of the feedback parameters (e.g., further processing is needed to determine the feedback parameters), the controller **78** may process the sensor data to determine the feedback parameters.

(33) In step **504**, one or more of the feedback parameters are monitored. The controller **78**, for example, may monitor the feedback parameter associated with the spatial position of the machining tool **36** and its tool head **72**. A change of the spatial position (e.g., downwards in FIG. **1**) over time corresponds to a feed rate of the tool head **72**; e.g., an estimated formation rate of the aperture **22**. Where this feed rate is outside of (e.g., greater than or less than) a (e.g., normal) threshold feed rate range, the control system **74** may determine the size of the tool head **72**. The sensor system **76**, for example, may measure the longitudinal length **80** of the tool head **72** and/or the lateral width **82** (e.g., diameter) of the tool head **72** and provide that measurement data to the controller **78**. The controller **78** may process this measurement data to determine the (e.g., actual) aperture formation rate. For example, a difference between the measured tool penetration depth (e.g., the aperture depth) and the longitudinal wear of the tool head **72** corresponds to the actual tool penetration depth. The controller **78** may process this actual tool penetration depth to determine the actual aperture formation rate.

(34) In step **506**, where the aperture formation rate is less than a formation rate threshold, the controller **78** may trigger a (e.g., adaptive) response. The controller **78**, for example, may signal the slurry delivery device **27** to adjust one or more slurry delivery parameters. For example, the controller **78** may signal the slurry delivery device **27** to increase a flowrate and/or a pressure of the slurry to the tool-workpiece interface **34**. The increased flowrate and/or pressure may increase the quantity of fresh slurry directed into the gap **35** between the tool tip **58** and the workpiece **24** as well as increase the outflow of the used slurry and the debris carried thereby from the gap **35** between the tool tip **58** and the workpiece **24**. This slurry replacement may increase a concentration of the abrasive particles within the slurry at the tool-workpiece interface **34** as well as replace dull abrasive particles with fresh sharp abrasive particles. The increase in the slurry flowrate may thereby increase machining efficiency and, thus, increase the aperture formation rate. A setpoint for the new increased flowrate of the slurry may be determined using a physics-based control model implemented by the controller **78**.

(35) In step **508**, the control system **74** continues to monitor the aperture formation rate in real time during the forming of the aperture **22**. Where the aperture formation rate is (or decreases) below the formation rate threshold (or another threshold), the slurry flowrate and/or pressure may be further increased. However, where the aperture formation rate is (or increases) a certain amount above the formation rate threshold (or another threshold), the slurry flowrate and/or pressure may be decreased. This process may be iteratively repeated during the formation of the aperture **22** until the aperture formation rate is within a desired range. The control system **74** may thereby implement automatic feedback control of the slurry delivery device **27** and flow of the slurry through the gap **35** between the tool tip **58** and the workpiece **24**.

(36) In some embodiments, where the aperture formation rate decreases below a second (e.g., minimum) formation rate threshold, the control system **74** may control the machining system components **27** and **28** to flush out the partially formed aperture in the workpiece **24**. For example, referring to FIG. **6**A, the tool holder **52** may remove the machining tool **36** from the partially formed aperture **22**′. While the machining tool **36** is removed, the slurry delivery device **27** may direct a flow of the slurry into the partially formed aperture to remove the used slurry as well as remove the workpiece debris that may have collected within the partially formed aperture. Referring to FIG. **6**B, the tool holder **52** may subsequently position the tool head **72** back into the partially formed aperture and the formation (e.g., machining) of the aperture **22** in the workpiece **24** may be resumed.
(37) In some embodiments, referring to FIGS. **7** and **8**, the machining tool **36** may be configured with an internal passage **90**; e.g., an inner bore. This internal passage **90** extends longitudinally within the machining tool **36** and its tool head **72** to an orifice **92** in the tool tip **58**. The internal passage **90** is configured to direct the slurry to and/or from the tool-workpiece interface **34**; see FIGS. **9**A and **9**B. For example, the internal passage **90** may be fluidly coupled with the source **38**. In such embodiments, referring to FIG. **9**A, the fresh slurry may be directed through the internal passage **90** to the tool-workpiece interface **34**. Here, the tool head **72** may also be configured as the nozzle **40**, or an additional nozzle. In another example, the internal passage **90** may be fluidly coupled with a vacuum device **94**. In such embodiments, referring to FIG. **9**B, the used slurry and the workpiece debris therewithin may be extracted out of the tool-workpiece interface **34** through the internal passage **90**. In both examples, the internal passage **90** may facilitate (a) flushing of the gap **35** of FIG. **2** between the tool tip **58** and the workpiece **24** and/or (b) the normal flow of the slurry through the gap **35** between the tool tip **58** and the workpiece **24**.
(38) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A method for machining a workpiece, comprising: forming an aperture in the workpiece using a machining system comprising an ultrasonic machining device, a slurry delivery device and a controller, the slurry delivery device comprising a passage that extends within the ultrasonic machining device to a tip of the ultrasonic machining device, the forming of the aperture comprising delivering a slurry to an interface between the ultrasonic machining device and the workpiece using the slurry delivery device, and transmitting ultrasonic vibrations into the slurry using the ultrasonic machining device; monitoring a feedback parameter during the forming of the aperture using the controller; and adaptively adjusting a slurry delivery parameter for the slurry delivery device during the forming of the aperture based on the feedback parameter using the controller; wherein the slurry is delivered to the interface through the passage during the forming of the aperture, and the slurry is removed from the interface through the passage during the forming of the aperture.
2. The method of claim 1, wherein the workpiece comprises a ceramic matrix composite material.
3. The method of claim 1, wherein the slurry comprises a plurality of abrasive particles within a carrier liquid.
4. The method of claim 3, wherein the plurality of abrasive particles comprise a carbide and/or

diamond.
5. The method of claim 1, wherein the slurry delivery parameter comprises a pressure of the slurry.
6. The method of claim 1, wherein the slurry delivery parameter comprises a flowrate of the slurry.
7. The method of claim 1, wherein the adjusting of the slurry delivery parameter initiates flushing out of the slurry at the interface by directing the slurry through the ultrasonic machining device.
8. The method of claim 7, wherein the slurry is pumped through the ultrasonic machining device to the interface.
9. The method of claim 7, wherein the slurry is drawn out from the interface into the ultrasonic machining device.
10. The method of claim 1, wherein the feedback parameter further comprises a load on the ultrasonic machining device.
11. The method of claim 1, wherein the feedback parameter further comprises a size of a tool of the ultrasonic machining device.
12. The method of claim 1, wherein the slurry delivery parameter is adjusted based on a physics-based model.
13. The method of claim 1, wherein the workpiece comprises a component of a gas turbine engine.
14. The method of claim 1, further comprising: providing a control signal based on the feedback parameter; and communicating the control signal to the slurry delivery device to adjust the feedback parameter.
15. The method of claim 1, further comprising adjusting the slurry delivery parameter to flush out the slurry at the interface by directing the slurry through the ultrasonic machining device.
16. The method of claim 1, wherein the controller automatically controls at least one of a flowrate or a pressure of the slurry to the workpiece.
17. The method of claim 1, wherein, during the forming of the aperture, the controller operates as a closed loop when adaptively adjusting based on the slurry delivery parameter.